

PyLIPS: an Open-Source Python Package to Expand Participation in Embodied Interaction

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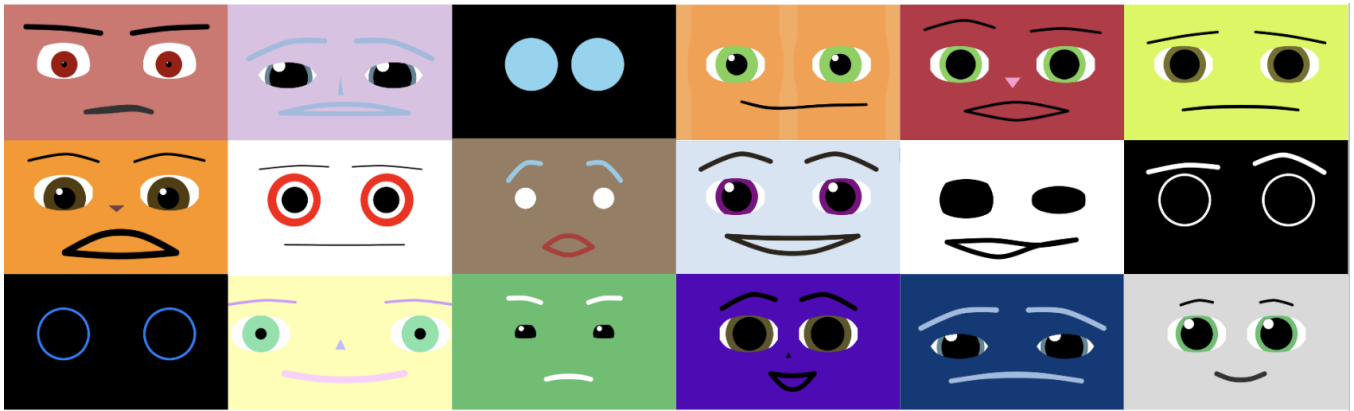


Figure 1: Samples from the design space of PyLIPS. PyLIPS enables users to easily develop expressive conversational agents with different levels of human-likeness. This package provides an interface to programmatically send speech and expression commands to an animated face.

Abstract

We demonstrate PyLIPS, a Python package for expanding access to screen-based facial interfaces for text-to-speech. PyLIPS can be used to rapidly develop social interactions for a wide variety of applications. We designed PyLIPS to be easy to use for novice users and expressive for experienced interaction designers. We demonstrate key features of PyLIPS: compatibility across devices, customizable face appearance, and automated lip synching for text inputs. PyLIPS can be found at <https://github.com/interaction-lab/PyLIPS>.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design; User interface programming.**

Keywords

Conversational Agents, Human-Robot Interaction, Human-Computer Interaction

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1 Introduction

The field of socially assistive robotics (SAR) [13, 21] has demonstrated that robots interacting with users by providing social support can enact positive behavior change in a wide variety of users. For example, social support from a robot can assist users with stroke to perform physical therapy exercises [9], users with ADHD symptoms to become acclimated to college environments [24], students with mild depressive symptoms to practice cognitive behavioral therapy exercises [17], and users with low vision to use magnifiers [38]. A major benefit of using socially assistive robots for these tasks is the robots' physical embodiment, which provides additional social modalities to motivate and assist the user [8, 11]. One of the

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most salient modalities in interaction are embodied facial cues, such as gaze and expression [1, 37].

Screen-based faces are widely adopted in robotics because they are versatile, increase social presence, and can provide flexibility to adapt to a user’s visual preferences [16]. Despite the widespread use of screen-based faces, few open-source software options exist for their development and deployment. Users are often required to purchase robots and learn to use proprietary software that does not transfer to other robot platforms [2, 3].

While some open-source software packages exist for conversational agents, they prioritize human-likeness [19]. In the case of SARs, anthropomorphic realism can be incongruent with the overall robot design and can increase user discomfort with the system [11]. *Our demonstration of PyLIPS fills a gap in the availability of stylized screen-based faces with text-to-speech capabilities.* PyLIPS is a Python package to easily implement social interactions with a screen-based face. We show the range of achievable designs and expressions in Figure 1.

2 Designing PyLIPS

Our design goal is to create a Python package that allows novice roboticists and conversational agent designers to easily develop facial interfaces. We focused on three design goals: simplicity, expressivity, and customizability.

2.1 Simplicity

Our primary goal was to develop open-source software that can be quickly learned to prototype and test different kinds of social interactions.

Python Implementation. We chose to implement the PyLIPS interface in Python, a popular programming language often taught in introductory coding classes. This makes PyLIPS accessible to novice users, while also maintaining usefulness for more experienced programmers. In addition, Python has strong community support [36] and tools for installing packages from the Python Package Index [14] allowing users to spend less time resolving package dependencies. Python is also a supported language for ROS [26], a broadly used robotics framework. This allows PyLIPS to easily integrate with existing workflows.

Cross-platform Browser-based Interaction. The PyLIPS face is displayed on a web browser, allowing cross-platform support; any platform that can open a browser can be used to run PyLIPS. This allows any computation for dialog or text-to-speech processing to occur off-board if the system running the PyLIPS face is working with limited computation. We note that although the face is displayed using a browser, the functionality of PyLIPS allows the package to be run offline. This can assist with developing interactions where web access is limited due to security concerns (e.g., with health data) or environmental concerns (e.g., remote outdoor interactions).

Minimal Hardware Requirements. The web interface uses scalable vector graphic (SVG) elements to render the face. There is a low computational demand to render SVG objects, and SVGs can scale to any screen size without losing resolution. These design choices allow the PyLIPS client to run on low-power computers, such as the Raspberry Pi [27] while maintaining high frame rates.

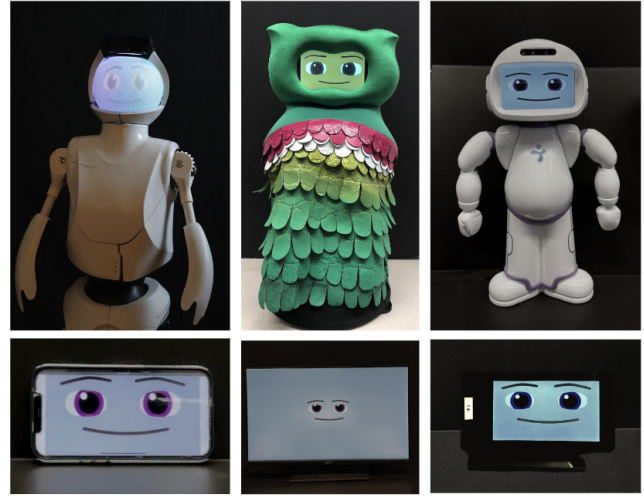


Figure 2: PyLIPS rendered on different robots and screens. The face can be adapted to different embodiments to match their external designs. Users can design their own robots with readily available screens, such as smartphones, laptops, and Raspberry Pi LCD displays.

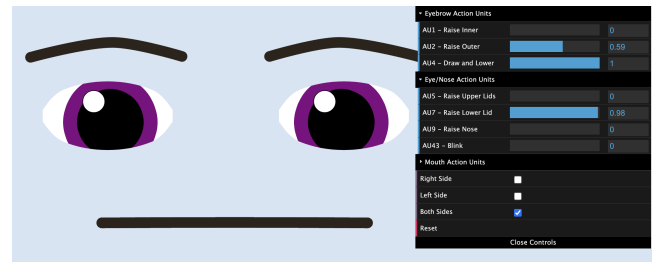


Figure 3: PyLIPS expression editor. Users can manually control different Action Units to create expressions.

We have demonstrated PyLIPS on several robot embodiments, including the Spritebot [30], Quori [31], and QTRobot [25]. We have also used different screens for PyLIPS, including smartphones, monitors, and LCD displays, as shown in Figure 2.

2.2 Expressivity

Action Unit Expressions. We allow users to directly modify the expressions of the face by specifying the AUs to activate, their activation level, and the time to move to that new AU configuration, as shown in Figure 3. Additionally, we allow users to specify a unilateral (i.e., one-sided) AU to create more dynamic expressions. By specifying AUs, users can generate validated expressions from the psychology literature, as well as experimental behaviors to accompany robot speech. AUs are also closely related to blendshapes, which are the standard interface for developing animated expressions [18].

Viseme-Based Lipsynching. We designed the face to use visemes based on the International Phonetic Alphabet [4], enabling lip-synching in different languages, since all human-producible sounds are represented in IPA. By allowing speech in multiple languages,



Figure 4: Users interacting with the PyLIPS face across different user studies. The face is deployed in studies that recruited users of varying ages, dis/abilities, ethnicities, and genders. The robot is engaged in tasks with varying social roles, objectives, and environments.

PyLIPS can be used in research with non-English speaking participants who are often not represented in research [15], thereby improving accessibility.

2.3 Customizability

Parameterized Face Appearance. We allow the user to modify a variety of features of the face to fit different robot embodiments. Each facial feature can have a color and size, and features can be removed or added for different use cases, screen sizes, or user preferences.

Multi-Face Support. Some interactions involve multiple conversational agents [28]. To support such interactions, we allow users to specify different face names. Named faces can then be individually controlled from the same Python script by specifying which face to send the commands to.

Customizable Voices. Different robot embodiments can suggest different expected voices in users [23]. Allowing users to choose voices that an agent uses has also been shown to lead to increased perceptions of quality, facilitating conversational interaction [29]. We allow users to specify different voices to use for the speech commands for each face.

3 Example Applications

PyLIPS has been used to develop several socially assistive robot applications [5–7, 9, 10, 12, 22, 32–35], shown in Figure 4. We demonstrate three interactions with PyLIPS to reflect our design goals.

Simplicity. To demonstrate PyLIPS’s simplicity, we show how it can interact with other Python packages to quickly develop an interaction in Python on a Raspberry Pi. Inspired by prior work that developed a socially assistive robot to practice exercises for children with cerebral palsy [12], we show how users can communicate with the robot through thumbs up and thumbs down gestures in a number guessing game. Throughout the game, the PyLIPS face describes the instructions, performs guesses, and provides feedback on the user’s gestures. We process the user’s gestures using mediapipe [20] and provide instructions and feedback through PyLIPS.

Expressivity. To show PyLIPS’s expressivity, we show how users can author expressions. We designed an interface to manually select the degree to which different action units are expressed. Thus users can select the specific action units associated with expressions

from the psychology and affective computing literature. We also allow a user to teleoperate the PyLIPS face using their own facial expressions in real time using a webcam. This allows users to record natural and expressive movements for the face.

Customizability. To show PyLIPS’s customizability, we show an example applications of different robot characters reading from a script. This shows how PyLIPS can communicate across devices, and how each device can have different voices, appearances, and personalities.

4 Discussion and Future Work

This demonstration introduces PyLIPS, an open-sourced Python package for developing screen-based faces. We provide functionality to generate automatic speech animations for arbitrary text, and to generate controlled facial expressions. We provide documentation to support novice users learning to program in Python at pylips.readthedocs.io. Our future work aims to investigate how social modalities of PyLIPS can be further automated to support designers; for example, through automatically generating expressions that match the sentiment of the text, or through developing consistent personas for each face to consistently deliver speech and expression.

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